

Multimedia Databases

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Contents

Equip learners with foundational knowledge of multimedia data organization and multimedia data processing.

Provide practical training in multimedia data processing techniques.

- Overview of multimedia databases
- Multimedia data management
- Multimedia information retrieval systems
- Indexing and data querying techniques

Textbooks

- [1] Đỗ Trung Tuấn, “Bài giảng Hệ CSDL đa phương tiện”, Nhà xuất bản bưu điện, Hà Nội, 2011.
- [2] Subrahmania V. S., “Principles of Multimedia Database Systems”, The Morgan Kaufmann Publisher, 1998.
- [3] Guojun Lu, “Multimedia Database Management Systems”, Artech House, 1999.
- [4] Bhavani Thuraisingham, “Manage and Mining Multimedia Databases”, CRC Press, 2001.

Grading

Types	Percentage	Objects
- Attendance	10%	Individuals
- Term project	90%	
Reports through email	$10\% + 20\% = 30\%$	Group
Demonstration (Testing date)	60%	Individuals

Term project

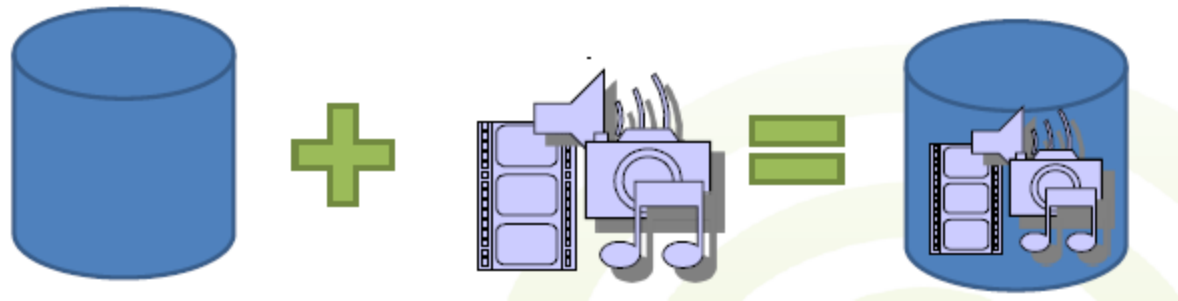
- Students work on assignments in groups of 2–3 members
- Develop a small multimedia database system
 - Data collection: acquire a thorough understanding of the characteristics of the collected data.
 - Feature extraction
 - MM Database and MM data queries

Today's content

- What multimedia databases store beyond text and numbers
- Why MMDBs need features, similarity search, and specialized indexes
- Real-world applications in surveillance, healthcare, education, and media
- How retrieval quality is evaluated with precision and recall
- Bridge to later lectures: data types, modeling, indexing, and retrieval pipelines
- **Goal:** connect introductory concepts to systems students already use every day

Introduction

- What are multimedia databases (MMDB)?
 - **Databases + multimedia = MMDB**



- Key words: **databases** and **multimedia**
- We already know databases, so what is **multimedia**?

Basic Definitions

- **Multimedia**

- The concept of **multimedia** expresses the integration of different digital media types
- The integration is usually performed in a document
- Basic media types are text, image, vector graphics, audio and video



Data types

- Text
 - Text data, Spreadsheets, E-mail, ...
- Image
 - Photos (Bitmaps), Vector graphics, CAD, ...
- Audio
 - Speech and music records, annotations, wave files, MIDI, MP3, ...
- Video
 - Dynamical image record, frame-sequences, MPEG, AVI, ...

What Makes an MMDB Different from a Traditional DB?

- Traditional DBs mainly answer exact-match queries on structured attributes
- MMDBs must manage large media objects plus descriptive metadata and feature vectors
- Queries often ask for similarity: 'find clips like this one' or 'images showing a sunset'
- Results are usually ranked rather than simply filtered
- Storage, indexing, and user interaction all become harder because meaning is not explicit
- This is why multimedia retrieval combines database, signal processing, and IR techniques

Multimedia Object Lifecycle

- Capture or collect the raw object: image, audio, video, or compound document
- Preprocess and normalize formats, duration, resolution, or sampling rate
- Extract metadata and features suitable for later retrieval
- Store raw media and derived descriptors in the database
- Run exact or similarity queries over indexed descriptors
- Present ranked results and optionally collect relevance feedback from the user

Documents

- Media objects may be text-only, image-only, audio-only, or video-only
- Multimedia documents combine several media into one logical object
- Earliest definition of information retrieval: “Documents are **logically interdependent** digitally encoded texts”
- Extension to multimedia documents allows the additional integration of **other media types** as images, audio or video



Documents

- Document types
 - **Media** objects are documents which are of **only one type** (not necessarily text)
 - **Multimedia** objects are general documents which allow an arbitrary **combination of different types**
 - Examples: lecture recording + slides, news article + photos, product page + demo video
- MMDB design must model both the raw object and relations among its parts
- This matters later for annotation, synchronization, and cross-media retrieval
- Multimedia data is transferred through the use of a **medium**

Basic Definitions

- **Medium**

- A medium is a carrier of information in a communication connection
- It is independent of the transported information
- The used medium can also be **changed** during information transfer

Example of Medium

- **Books**



- Communication between authors and readers
- Independent from content
- Hierarchically built on text and images
- Reading out loud represents medium change to sound/audio

Media classification

- Based on **receiver type**
 - Visual/optical medium
 - Acoustic mediums
 - Haptical medium – through tactile senses
 - Olfactory medium – through smell
 - Gustatory medium – through taste
- Based on **time**
 - Dynamic
 - Static

Why Multimedia Data Is Hard to Manage

- Media objects are large, heterogeneous, and often time-dependent
- Two files with the same meaning may have very different raw byte patterns
- Users usually search by concept, while systems start from low-level signals
- Retrieval must balance response time, storage cost, and usefulness of results
- Compression, indexing, annotation, and device adaptation are built-in requirements
- MMDBs therefore extend both storage management and information retrieval ideas

Multimedia databases

- Why do we need **databases**?
 - Most important operations of databases are **data storage** and **data retrieval**

Multimedia Databases

- **Persistent storage** of multimedia data, e.g.:
 - Text documents
 - Vector graphics, CAD
 - Images, audio, video
- **Content-based retrieval**
 - Efficient content-based search
 - Standardization of meta-data.

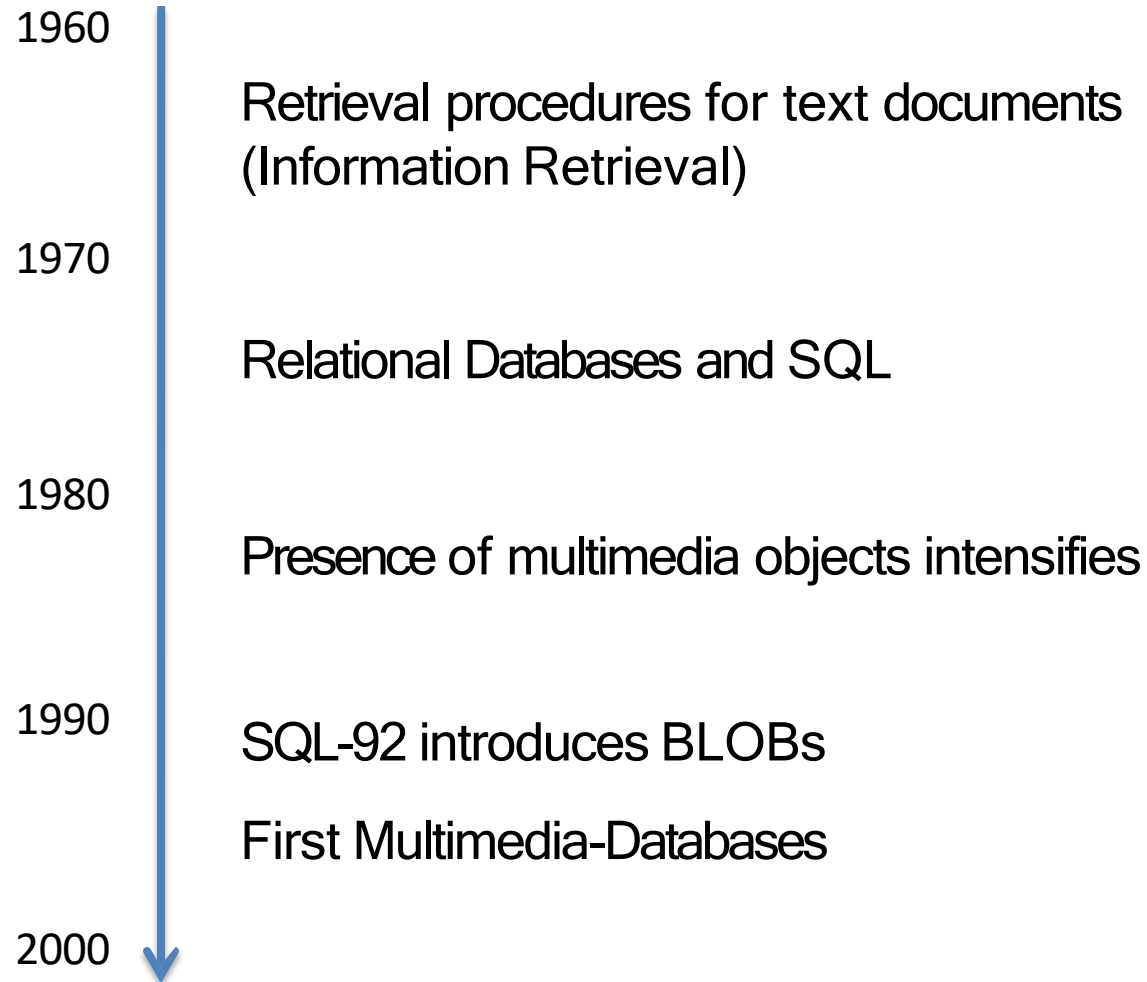
Multimedia Databases

- **Stand-alone vs. database storage model?**
 - Special retrieval functionality as well as corresponding optimization can be provided in both cases...
 - But in the second case we also get the general advantages of databases
 - Declarative query language
 - Orthogonal combination of the query functionality
 - Query optimization, Index structures
 - Transaction management, recovery
 - ...

Storage Model: Raw Objects plus Searchable Descriptors

- Keep the original media file for playback, display, or legal archiving
- Store metadata such as title, author, location, timestamp, and format
- Store derived features such as color histograms, MFCCs, keyframes, or embeddings
- Indexes are usually built on metadata and features, not directly on raw bytes
- This hybrid model supports both standard SQL-style filtering and content-based search
- It also allows updates to feature pipelines without losing the master media object

Historical Review



Requirements

- **Requirements** for multimedia databases (Christodoulakis, 1985)
 - Classical **database functionality**
 - **Maintenance** of unformatted data
 - Consideration of special **storage** and presentation devices

Requirements

- To comply with these requirements the following aspects need to be considered
 - **Software architecture** –new or extension of existing databases?
 - **Content addressing** – identification of the objects through content-based features
 - **Performance** – improvements using indexes, optimization ...

Requirements

- **User interface** – how should the user interact with the system?
Separate structure from content!
- **Information extraction** – (automatic) generation of content-based features
- **Storage devices** –very large storage capacity, redundancy control and compression
- **Information retrieval** – integration of some extended search functionality

Data Retrieval

- **Retrieval:** choosing between data objects. Based on ...
 - a SELECT condition (exact match)
 - or a defined similarity connection (best match)
- Retrieval may also cover the **delivery** of the results to the user

Data Retrieval

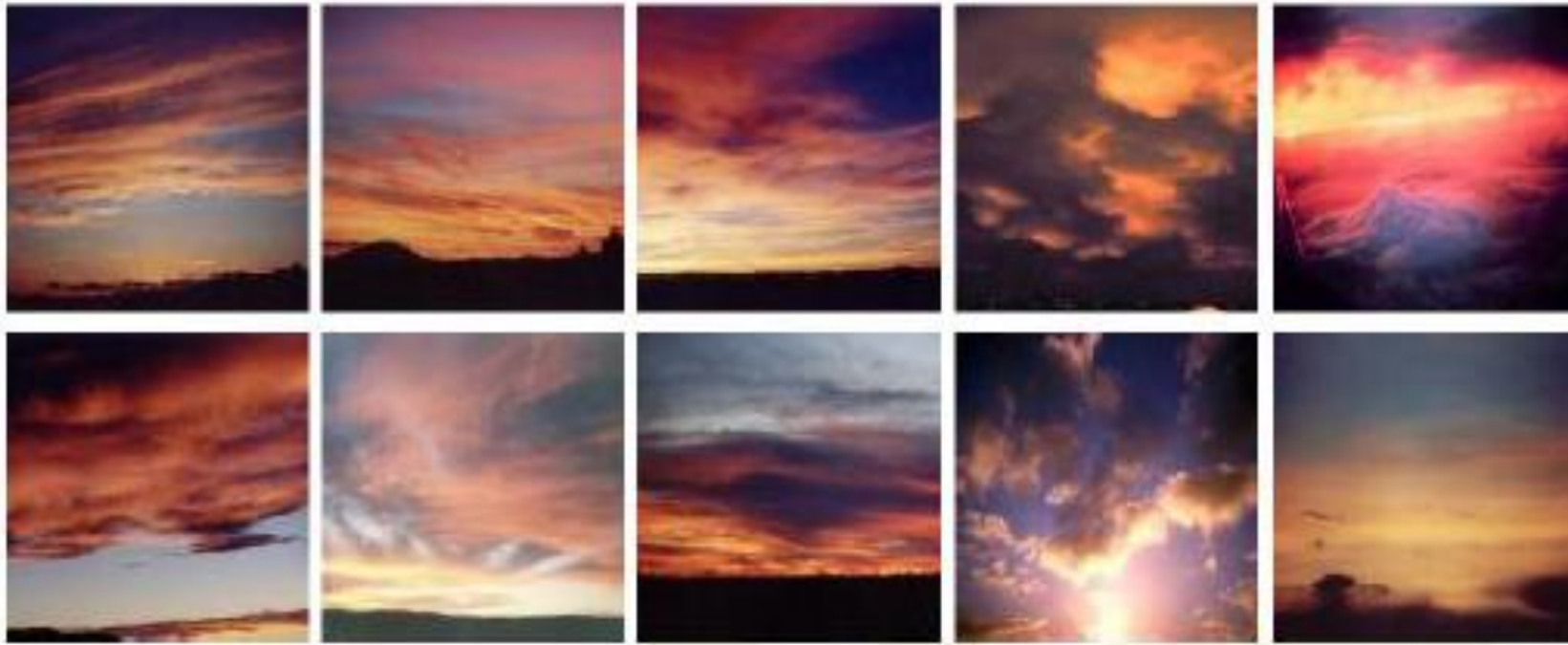
- Closer look at the **search functionality**
 - “Semantic” search functionality
 - Orthogonal integration of classical and extended functionality
 - Search does not directly access the media objects
 - Extraction, normalization and indexing of content- based features
 - Meaningful similarity/distance measures

Exact Match vs Similarity Retrieval

- Exact match asks for an object satisfying a strict condition on attributes
- Similarity search compares descriptor vectors and returns the nearest matches
- Exact match fits metadata such as date, author, or file format
- Similarity search fits content questions such as 'videos like this clip'
- Most practical MMDB systems combine both in a single query workflow
- Ranking is essential because relevance is often gradual, not binary

Content-based Retrieval

- “Retrieve all images showing a sunset !”



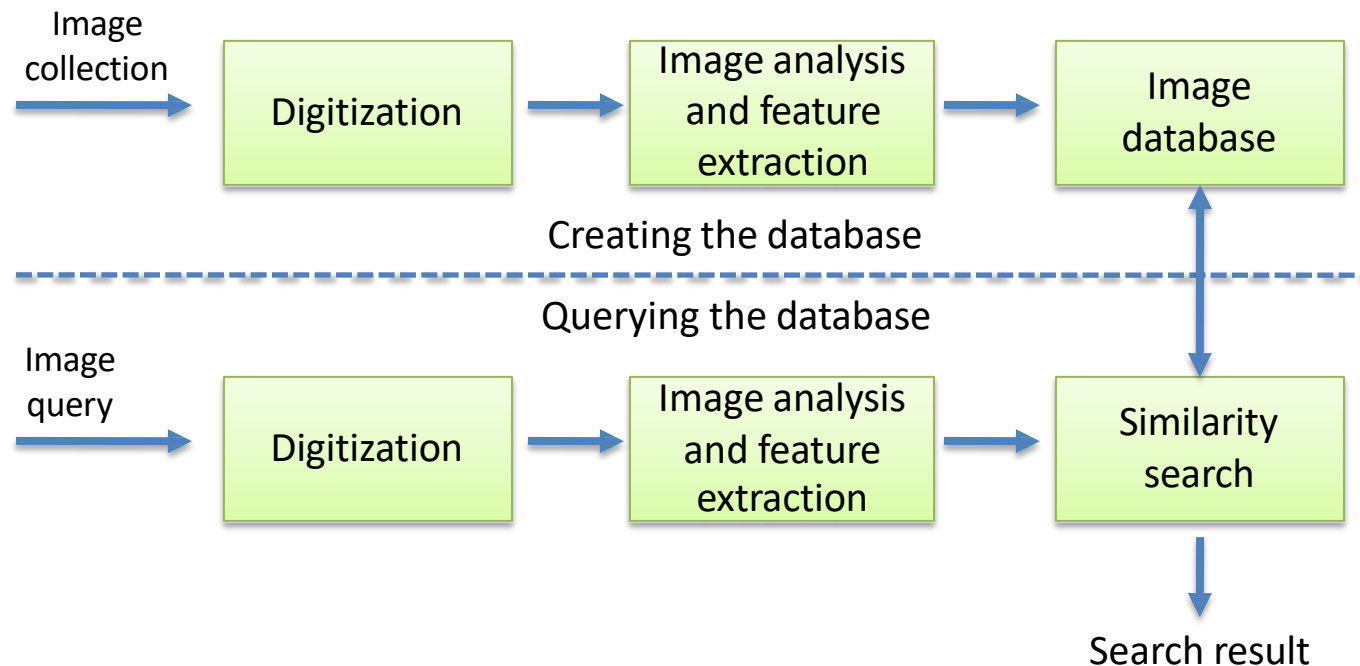
- What exactly do these images have in common

Semantic Retrieval — Introduction

- Users want results matching meaning, not just file names or manual tags
- Semantic retrieval tries to connect content descriptors with higher-level concepts
- Typical pipeline: preprocessing, feature extraction, indexing, similarity search
- Annotations, learned models, and feedback help narrow the meaning gap
- This is the motivation behind the following architecture slides

Semantic View

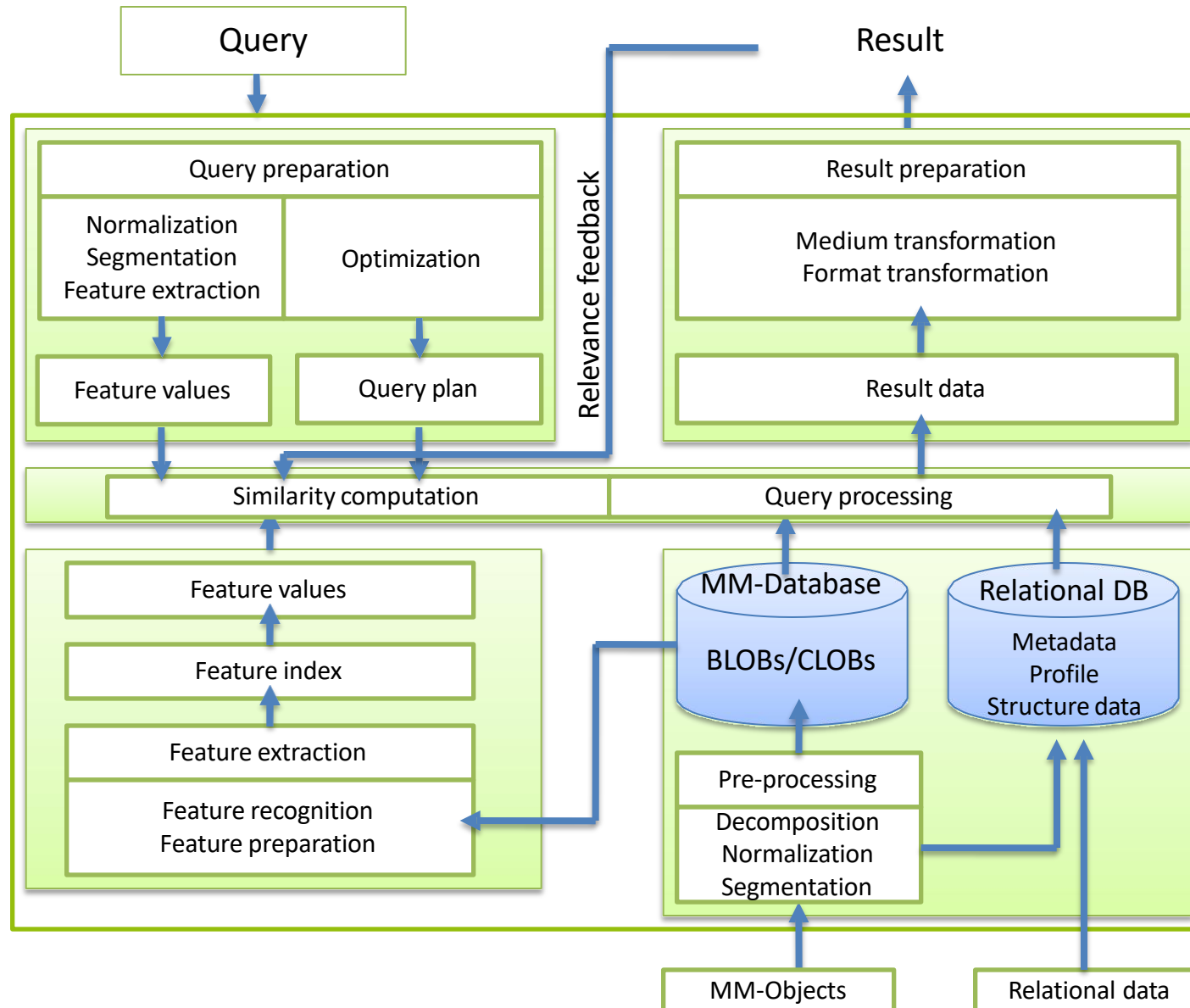
- Usually 2 main steps
 - Example: image databases



- Data retrieval



More detailed View



Semantic Search and the Meaning Gap

- Users think in concepts such as sunset, lecture, accident, or suspicious behavior
- Systems often start from low-level features such as color, texture, motion, or pitch
- The mismatch between machine descriptors and human meaning is the semantic gap
- Metadata, learned embeddings, ontologies, and relevance feedback help reduce this gap
- A practical MMDB rarely depends on only one source of evidence
- This topic connects directly to later lectures on metadata and multimedia IR

Content-Based Retrieval Workflow

- Offline stage: ingest media, normalize it, extract descriptors, and build indexes
- Online stage: process the query object, extract the same descriptors, and compare
- Similarity scores are computed against candidate objects from the repository
- The system ranks results, prepares previews, and may support feedback/refinement
- Key engineering choices: descriptors, distance measure, and candidate-pruning strategy
- The architecture on slides 26–28 becomes much clearer when seen as an offline/online split

Applications

- Lots of multimedia content on the Web
 - Social networking e.g., Facebook, MySpace, Hi5, Orkut, etc.
 - Photo sharing e.g., Flickr, Photobucket, Imeem, Picasa, etc.
 - Video sharing e.g., YouTube, Megavideo, Metacafe, blip.tv, Liveleak, etc.



Real Application: Streaming and Media Platforms

- Netflix, Spotify, and TikTok all manage massive multimedia catalogs
- Metadata supports filtering by genre, creator, date, and language
- Content descriptors support near-duplicate detection, thumbnail selection, and recommendation
- Speech-to-text enables search inside podcasts, lectures, and subtitles
- User behavior adds another signal, but the MMDB still stores and serves the core media objects
- This is a modern large-scale example of MMDB ideas from the lecture

Semantic View – Image data

- Database creation: collect images, digitize if needed, then extract descriptors
- Query phase: analyze the input image the same way as database objects
- Similarity search compares query descriptors with stored descriptors
- Results are ranked and returned to the user as best matches
- The same pattern also applies to audio and video retrieval systems

Real Application: Smart-City Surveillance

- Cities store continuous streams from street, traffic, and building cameras
- Metadata includes camera ID, location, timestamp, and event type
- Content features help detect vehicles, faces, unusual motion, or crowd density
- Operators rarely browse raw archives sequentially; they retrieve by event or similarity
- This application explains why 'cameras are everywhere' matters as a database problem
- Latency, retention policy, and privacy become major design constraints

Applications

- **Cameras** are everywhere

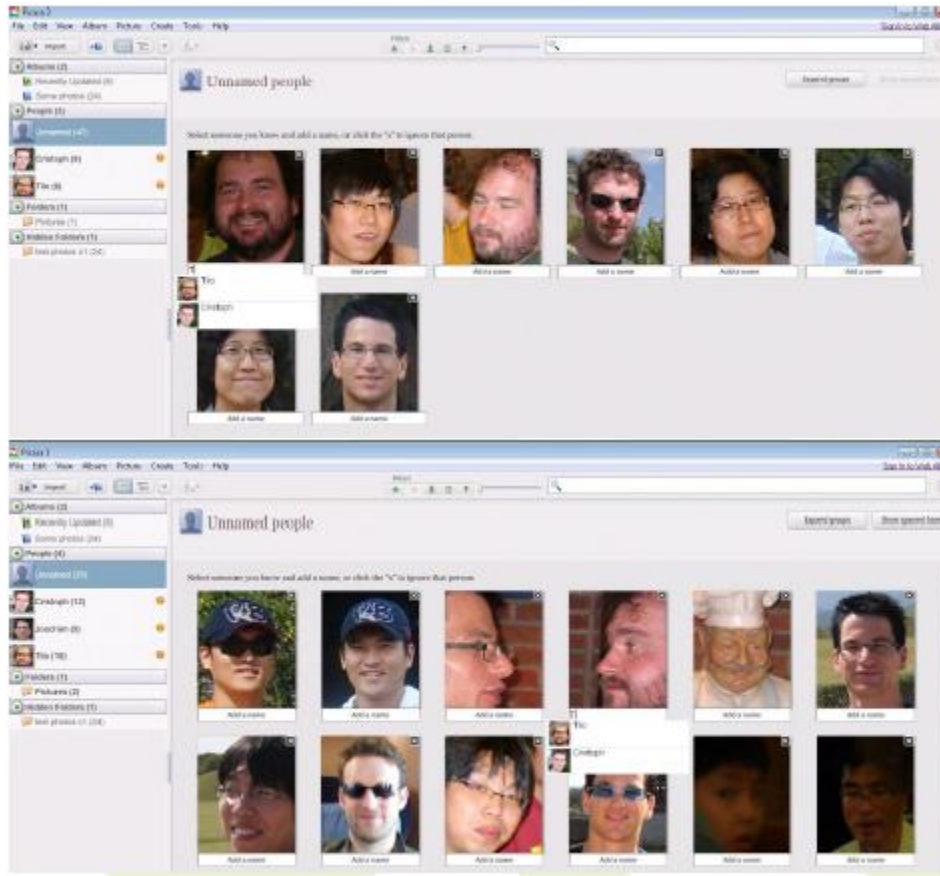


Applications — Cameras Are Everywhere

- Phones, traffic cameras, stores, classrooms, and homes continuously create visual data
- The database challenge is not just storage volume but also fast event retrieval
- Practical queries: find same vehicle, similar face, crowd incident, or suspicious motion
- This is why multimedia archives need descriptors and indexes, not only filenames

Applications

- face recognition



Real Application: Face Search and Access Control

- Face recognition systems keep image templates or embeddings, not only photos
- A query image is matched against stored descriptors and ranked by similarity
- Typical uses: device unlock, airport screening, attendance, and missing-person search
- False positives and false negatives have different costs depending on the application
- This slide naturally motivates the evaluation section later in the lecture
- It also shows why benchmark quality and threshold selection matter

Real Application: Medical and Scientific Archives

- Hospitals manage X-ray, CT, MRI, ultrasound, pathology images, and clinical video
- Researchers need retrieval by anatomy, abnormality pattern, study date, or modality
- Metadata alone is often insufficient, so image/video descriptors are important
- Audit trails and high-fidelity storage are critical because data affect patient care
- MMDB techniques also support microscopy, astronomy, and remote-sensing repositories
- This is one of the strongest real-world motivations for high-quality retrieval

Real Application: Digital Libraries and E-Learning

- Universities store lecture videos, slide decks, scanned books, and recorded seminars
- A single course object may combine text, image, audio, and video in one package
- Search can target speaker, keyword, slide image, or timestamped transcript segment
- Students care about relevance and navigation, not only whether a file exists
- This is a practical MMDB case students already experience as users
- It also illustrates compound documents and cross-media links

Real Application: E-Commerce and Brand Monitoring

- Online marketplaces store product photos, short videos, logos, and user reviews
- Retrieval supports similar-product search, duplicate listing detection, and moderation
- Visual search lets users upload a picture and find comparable items
- Brand owners search for copied media, counterfeit goods, and reused assets
- Metadata and content descriptors must work together for good precision
- The business value is immediate: conversion, trust, and policy enforcement

Sample Scenario

- Consider a **police investigation** of a large-scale drug operation
- Possible generated data:
 - **Video** data captured by surveillance cameras
 - **Audio** data captured
 - **Image** data consisting of still photographs taken by investigators
 - **Structured relational data** containing background information
 - **Geographic information system data**

Characteristics

- MMDB characteristics:
 - **Static:** high number of search queries (read access), few modifications of the data
 - **Dynamic:** often modifications of the data
 - **Active:** the functionality of the database leads to operations at application level
 - **Passive:** database reacts only at requests from outside
 - **Standard search:** queries are answered through the use of metadata e.g., Google-image search
 - **Retrieval functionality:** content-based search on the multimedia repository e.g., Picasa face recognition

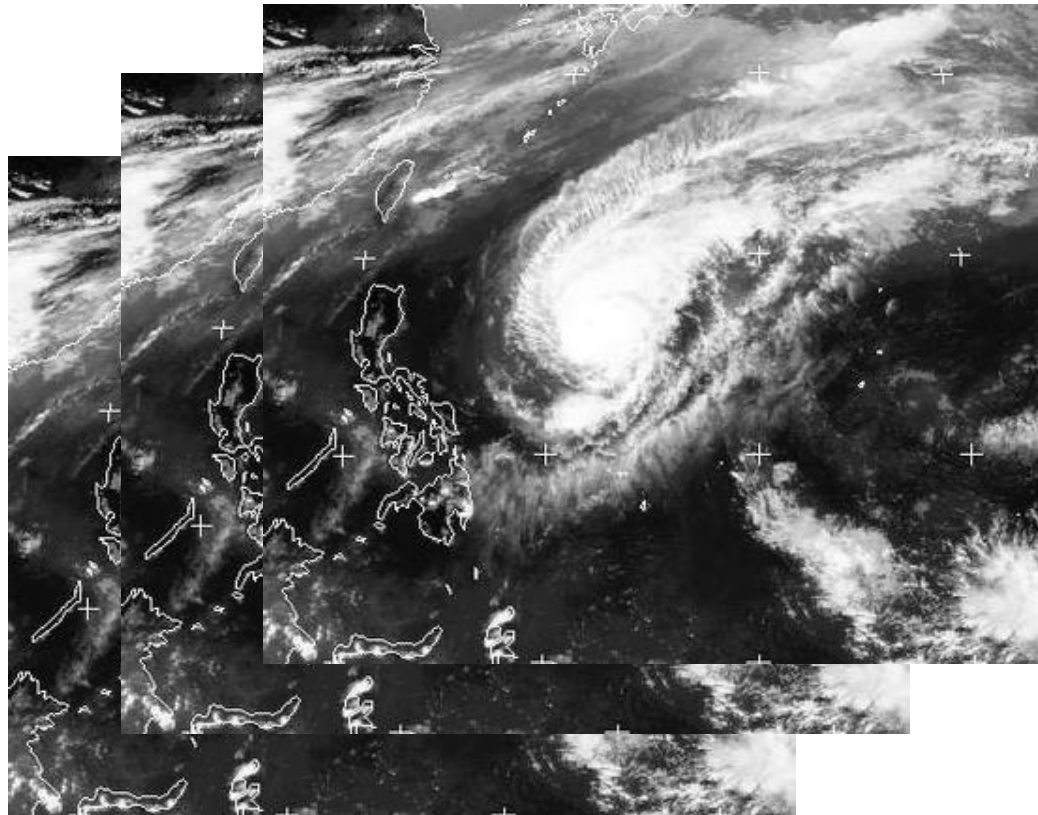
Example

- **Passive static retrieval**
 - Art historical use case



Example

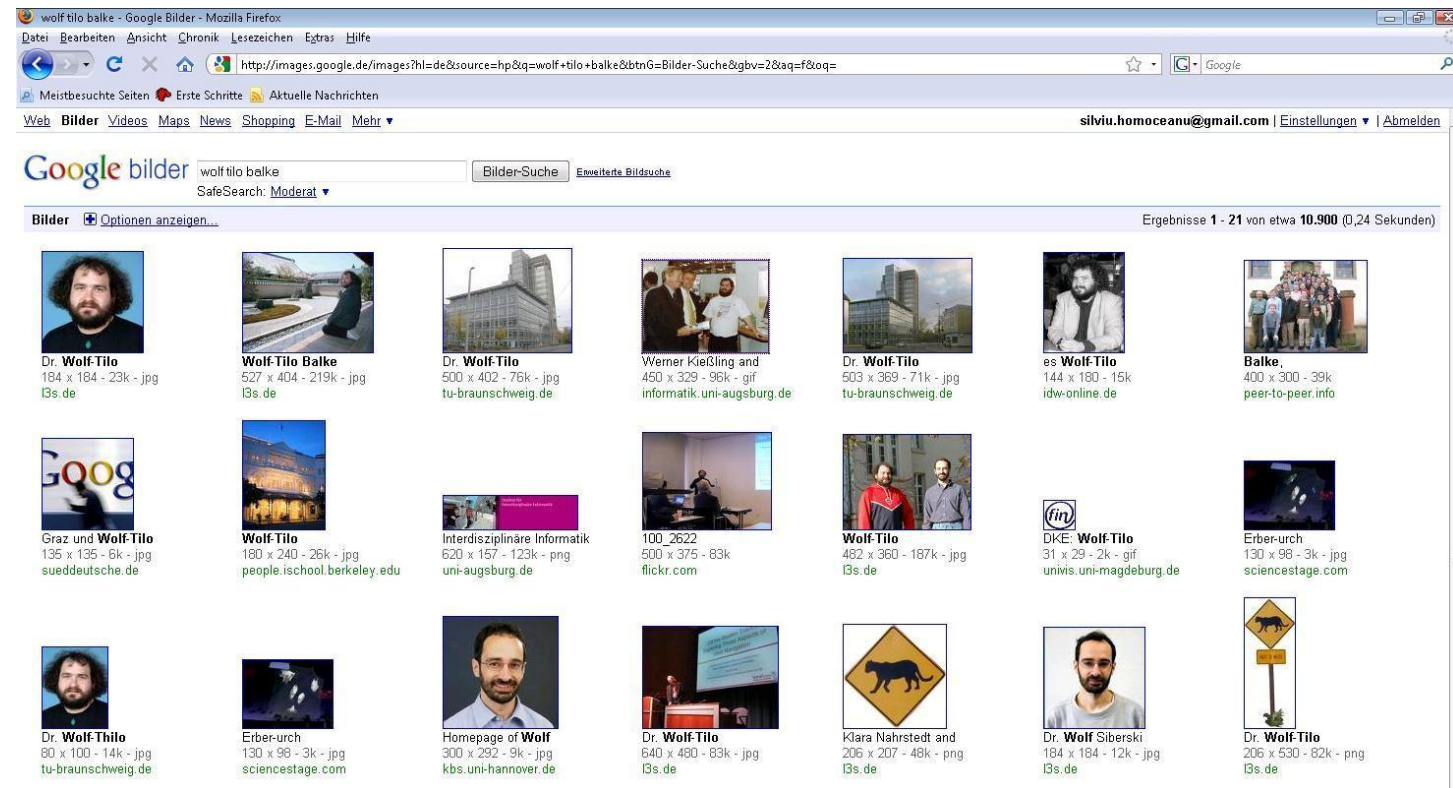
- **Active dynamic retrieval**
 - – Wetter warning through evaluation of satellite photos



**Typhoon-Warning
for the Philippines**

Example

- **Standard search**
 - Queries are answered through the use of metadata e.g., Google-image search



Retrieval Evaluation

- Basic **evaluation** of retrieval techniques
 - **Efficiency** of the system
 - Efficient utilization of system resources
 - Scalable also over big collections
 - **Effectivity** of the retrieval process
 - High quality of the result
 - Meaningful usage of the system
- What is more important? An effective retrieval process or an efficient one?
- **Depends on the application!**

Evaluation Goals for Multimedia Retrieval

- Efficiency asks whether the system is fast enough and uses resources well
- Effectiveness asks whether the returned results are actually useful for the query
- The right balance depends on the application: emergency response differs from casual browsing
- A very fast but irrelevant system is not successful; neither is a perfect but unusable one
- Real deployments usually measure both categories continuously

Evaluating Efficiency

- Characteristic values to measure
- **efficiency** are e.g.:
 - Memory usage
 - CPU-time
 - Number of I/O-Operations
 - Response time
- Depends on the (Hardware-) environment
- **Goal:** the system should be **efficient enough!**

Evaluating Effectivity

- Measuring **effectivity** is more difficult and always depending on the query
- We need to define some **query-dependent**
- evaluation measures!
 - Objective quality metrics
 - Independent from the querying interface and the retrieval procedure
 - Allows for comparing different systems/algorithms

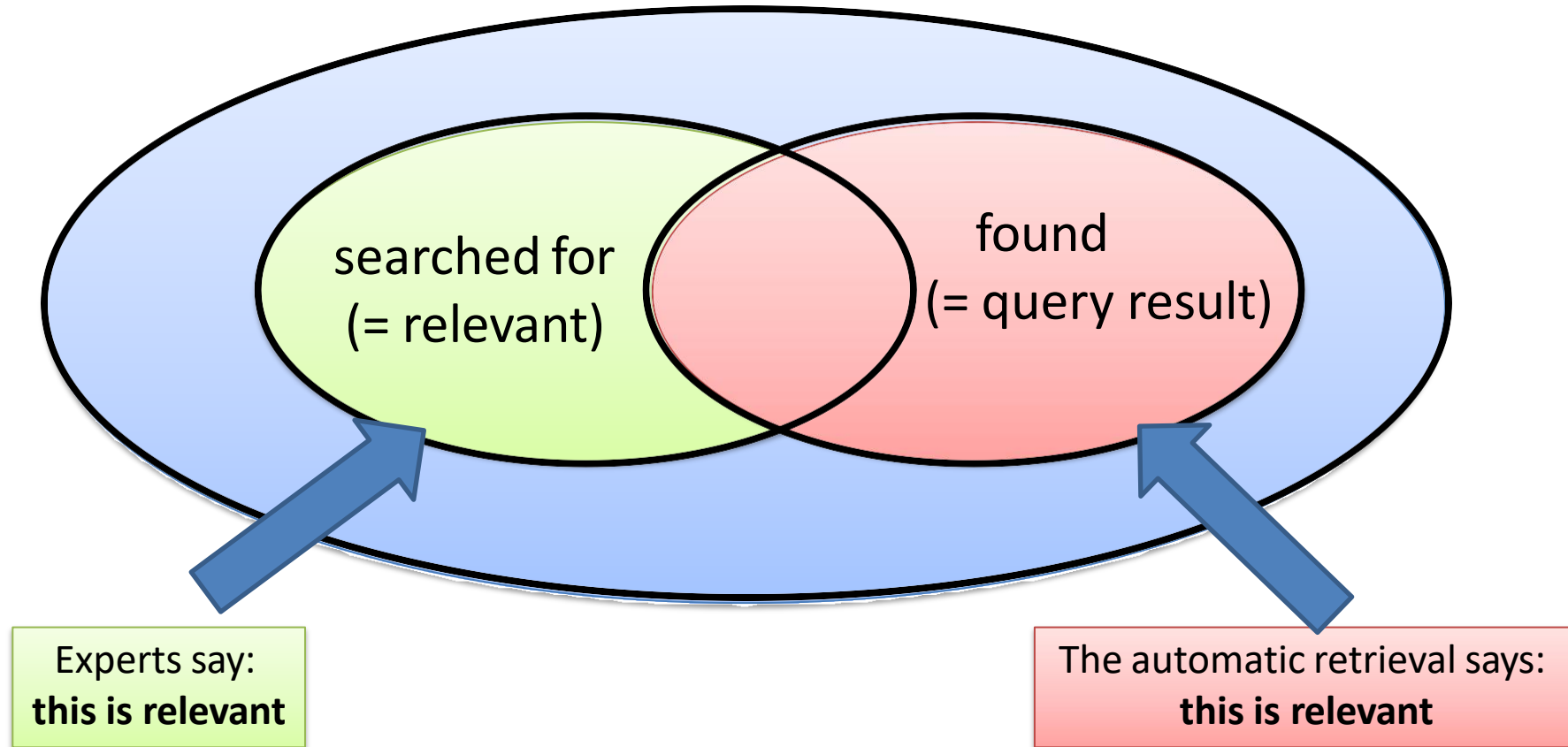
Evaluating Effectivity

- Effectivity can be measured regarding an
 - **explicit query**
 - Main focus on evaluating the behavior of the system with respect to a query
 - **Relevance** of the result set
 - But effectivity also needs to consider **implicit information needs**
 - Main focus on evaluating the usefulness, usability and user friendliness of the system

Relevance

- **Relevance** as a measure for **retrieval**: each document will be binary classified as **relevant** or **irrelevant** with respect to the query
 - This classification is manually performed by “**experts**”
 - The response of the system to the query will be compared to this classification
 - Compare the obtained response with the “ideal” result

Involved Sets



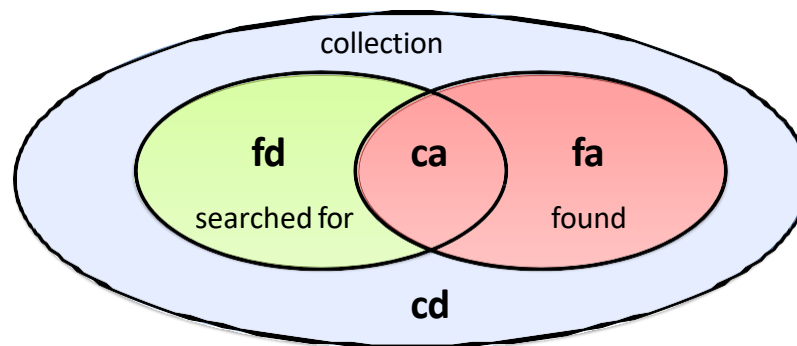
Evaluation

- Confusion matrix

System- evaluation User- evaluation	relevant	irrelevant
	relevant	irrelevant
relevant	ca 	fd  
irrelevant	fa 	cd 

False Negatives

- False negatives: relevant documents classified by the system as irrelevant
 - False dismissals
- Dangerous, since they **can't be detected easily by the user**
 - Are there “better” documents in the collection which the system didn't return?
 - False alarms are usually not as bad as false dismissals



Precision, Recall, and F1 in Practice

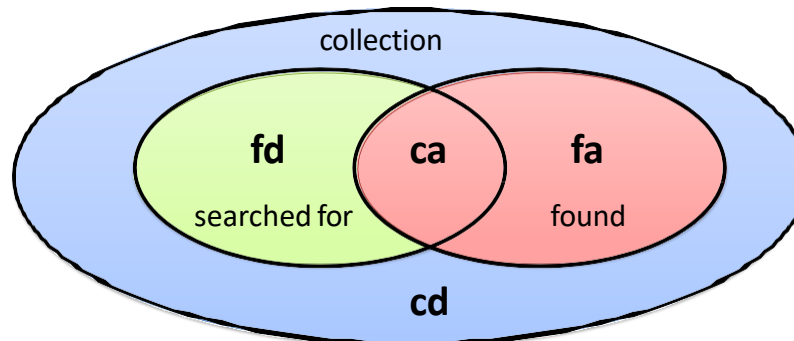
- Precision = relevant returned / all returned; useful when wrong alarms are costly
- Recall = relevant returned / all relevant; useful when missing an item is costly
- F1-score balances precision and recall into one summary value
- Face search, medical retrieval, and police investigation may prefer different trade-offs
- Thresholds, result set size, and re-ranking all change these numbers
- These metrics become meaningful only with a labeled benchmark or expert judgments

Benchmarking and User-Centered Evaluation

- Offline benchmarks compare systems using fixed queries and relevance labels
- Online evaluation studies usability, satisfaction, and whether tasks are completed faster
- Experts may label relevance differently from end users, so context matters
- A practical MMDB often needs both algorithmic scores and user-study evidence
- This explains why recall is harder to estimate than precision in the wild
- Slides 42–51 fit into this broader view of system evaluation

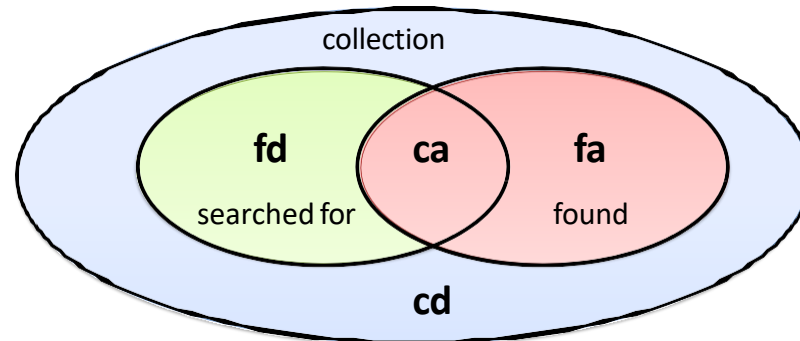
Remaining Sets

- Correct positives (correct alarms)
 - All documents correctly classified by the system as relevant
- Correct negatives (correct dismissals)
 - All documents correctly classified by the system as irrelevant
- All sets are disjunctive and their reunion is the entire document collection



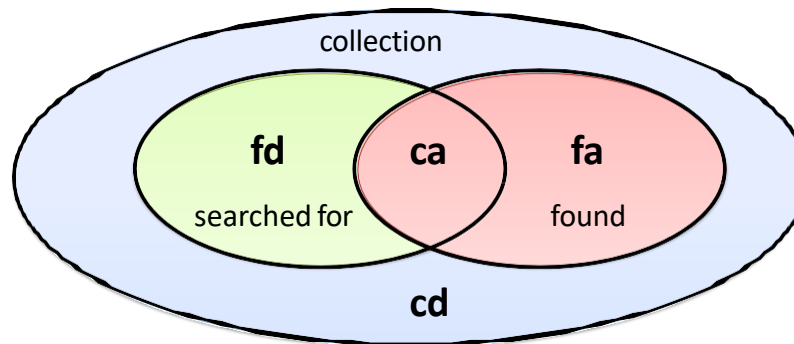
Interpretation

- Relevant results = $fd + ca$
 - Handpicked by experts!
- Retrieved results = $ca + fa$
 - Retrieved by the system



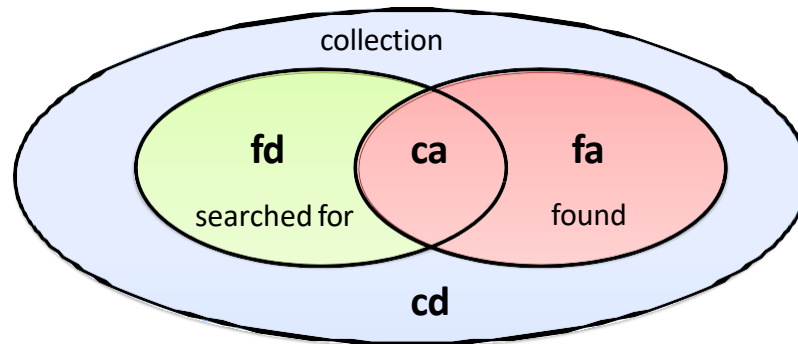
Precision

- **Precision** measures the ratio of correctly returned documents relative to all returned documents
 - $P = ca / (ca + fa)$
- Value between [0, 1] (1 represents the best value)
- High number of false alarms mean worse results



Recall

- **Recall** measures the ratio of correctly returned documents relative to all relevant documents
 - $R = ca / (ca + fd)$
- Value between [0, 1] (1 represents the best value)
- High number of false drops mean worse results



Precision-Recall Analysis

- Both measures only make sense, if considered **at the same time**
 - E.g., get perfect recall by simply returning all documents, but then the precision is extremely low...
- Can be balanced by tuning the system
 - E.g., smaller result sets lead to better precision rates at the cost of recall
- Usually the average precision-recall of more queries is considered (macro evaluation)

Actual Evaluation

- **Alarms** (returned elements) divided in *ca* and *fa*
 - Precision is easy to calculate
- **Dismissals** (not returned elements) are not so trivial to divide in *cd* und *fd*, because the entire collection has to be classified
 - Recall is difficult to calculate
- **Standardized Benchmarks**
 - Provided connections and queries
 - Annotated result sets

Summary: Why Multimedia Databases Matter

- MMDBs combine storage, metadata, feature extraction, indexing, and ranked retrieval
- They are necessary whenever users need to search inside large media collections by meaning
- Real applications span streaming, surveillance, medicine, libraries, and e-commerce
- Good systems combine exact-match metadata search with content-based similarity search
- Evaluation must consider both efficiency and usefulness of the results
- Later lectures deepen these ideas for data types, modeling, indexing, and retrieval